

# SIGNAGEHUB MODULAR LAYOUT ENGINE: ENTERPRISE PLUGIN INVENTORY CATALOG

This master architectural index outlines the complete suite of viable plugins for the SignageHub platform. In strict accordance with our offline-stable security architecture, **all direct external API calls are mediated by the SignageHub App Server**. The frontend `plugin.js` executes lightweight, asynchronous fetch calls strictly to our internal `api.signagehub.tv/v1/screen-data` endpoint on a 5-to-10 minute loop, caching states locally via IndexedDB (`localStorage`) to guarantee 24/7 stability.

All modules conform to the unbreakable 3-column matrix (`grid-template-columns: repeat(3, minmax(0, 1fr))`) and dynamically apply edge-anchoring `transform-origin` parameters for seamless 8K to 1080p hardware scaling.

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## CATEGORY 1: HOSPITALITY & CULINARY (Fine Dining, Bars & Luxury Hotels)

*Focus: Transactional and physical telemetry that elevates the guest experience without consumer-grade visual noise.*

### 1. The Sommelier Allocation HUD

- **Enterprise Data Source:** Backend ingests Toast POS Stock API (`status: OUT_OF_STOCK` or `QUANTITY`) and Oracle MICROS Symphony BI API/Transaction Services Gen2 with WISK variance analysis.
- **3-Column Grid Behavior:** Anchors to the top row (`transform-origin: top`). Maintains rigid 1/3 column geometry. When stock dips below the backend threshold, standard metadata fades via GPU-accelerated opacity transitions, replaced by an elegant `[ ALLOCATION: ONLY X BOTTLES REMAINING ]` text node. Uses warm brass "candle-glow" CSS pseudo-element animations in *Luxury Hotel Mode*.

### 2. Intelligent "Sold Out" Menu Masking

- **Enterprise Data Source:** Backend POS item availability webhooks (Lightspeed, Toast, Trivec) resolving to a boolean stock flag.
- **3-Column Grid Behavior:** Operates in the middle grid row (`transform-origin: center`). To prevent layout reflows and preserve symmetry, the item is never removed from the DOM. Instead, the container drops to 20% opacity and applies a 1px cross-strikethrough with the stamp `[ SOLD OUT – RETURNING FOR TOMORROW'S MENU ]`. In *Minimal Zen Mode*, this is rendered in ultra-thin, low-luminance white fonts against a matte transparent background.

### 3. Live Bar & Cellar Telemetry

- **Enterprise Data Source:** Backend parses local BACnet IP / UDP traffic from environmental sensors and localized temperature controllers/wine dispensers.
- **3-Column Grid Behavior:** Resides in a 1/3 column slot. Renders cellar conditions (Temp, Humidity, Draft Line pressure) as a continuous, slowly plotting Cartesian line chart via client-side canvas. Converts physical maintenance variables into monochromatic vectors that update elegantly every 5 minutes.

### 4. Dynamic Premium Happy Hour/Infusion Countdown

- **Enterprise Data Source:** POS scheduled events and catering order APIs synced through the App Server.

- **3-Column Grid Behavior:** Functions as an ambient spatial tracker in the bottom row (transform-origin: bottom). Instead of ticking digital clocks, it utilizes a subtle glassmorphic progress bar or a glowing border fill that advances at an ultra-low refresh rate to signify the readiness of barrel-aged cocktails or limited-time offerings.
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## CATEGORY 2: TRANSIT, INFRASTRUCTURE & CONCIERGE (Mobility Hubs & Lobbies)

Focus: Real-time spatial awareness and seamless departures.

### 1. Connoisseur Flight Board

- **Enterprise Data Source:** FlightAware AeroAPI (/airports/{id}/flights/scheduled\_arrivals), targeting specific tail numbers and local FBOs, enhanced by FlightAware Foresight predictive ML models.
- **3-Column Grid Behavior:** Enforces strict structural separation across the grid: Left (Origin/Destination), Center (Tail Number/Airline), Right (Status). In *Airport Flip Mode*, it uses fixed tile sizes to mimic mechanical split-flaps. In *Mission Control Mode*, it utilizes clean monospace typography and glowing vector paths.

### 2. Regional Transit Matrix

- **Enterprise Data Source:** Backend aggregator parsing GTFS Realtime feeds with a fallback to local GTFS Static zip packages.
- **3-Column Grid Behavior:** Fits into the middle row matrix. Uses color-blocked tags (e.g., #00AEEF) for specific transit lines. Layout changes are restricted to GPU-safe opacity fades—delayed trains trigger a subtle amber pulse instead of aggressive flashing red warnings. Perfect for the *Metro Transit Mode* aesthetic.

### 3. VIP Chauffeur & Rideshare Status Board

- **Enterprise Data Source:** Uber/Lyft Business APIs and private fleet GPS telematics via backend webhooks.
- **3-Column Grid Behavior:** Anchors to the bottom grid. Displays incoming fleet ETAs as monochromatic grey vectors with minimal highlights. A small, high-luminance amber dot represents active movement, crawling horizontally across the 1/3 column width.

### 4. Hotel Concierge Luggage/Shuttle Telemetry

- **Enterprise Data Source:** RFID baggage sorting database webhooks and internal shuttle GPS beacons.
  - **3-Column Grid Behavior:** Occupies a side column. Tracks baggage transitions from cargo holds to the lobby using a simplified, non-topographical vector projection line. Smooth translate3d CSS transforms govern the slow crawl of the data points.
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## CATEGORY 3: CORPORATE TELEMETRY & FINTECH (Executive Boardrooms & Banks)

Focus: Operational velocity, resource optimization, and market states.

### 1. Executive Sanctuary HUD

- **Enterprise Data Source:** Microsoft Graph API (findMeetingTimes, getSchedule) mapped to availabilityView blocks, plus Crestron (GLS-ODT-C-NS) BACnet IP occupancy and Airzone HVAC sensors.

- **3-Column Grid Behavior:** Center grid. Displays schedule states (Free/Busy/Tentative) via monochromatic glowing horizontal trace lines. HVAC airflow velocity is represented as a clean circular arc. Integrates "Walk-Through" and "Grace Occupancy" timing loops to prevent layout flickering.

## 2. Enterprise Velocity Ticker

- **Enterprise Data Source:** Internal corporate KPIs, server deployment status, and carbon offset milestones ingested by the backend.
- **3-Column Grid Behavior:** Bottom row continuous loop. Uses `translate3d` and `requestAnimationFrame` to crawl data at 1 pixel per frame. In *Executive Boardroom Mode*, it renders with sharp white typography on dark graphite with zero glow.

## 3. Global Market & Commodity HUD

- **Enterprise Data Source:** Financial REST API endpoints (e.g., Bloomberg, Alpha Vantage) normalized into standard JSON via the App Server.
- **3-Column Grid Behavior:** Top or Middle rows. Eliminates distracting UI badges. In *Trading Floor Mode*, it utilizes 50ms millisecond-sweeps to reflect market ticks, paired with restrained red/green coordinate markers.

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## CATEGORY 4: NEWS, MEDIA & GLOBAL AWARENESS (Public Spaces & Reception Desks)

*Focus: Content curation and informational streams that keep environments connected.*

### 1. Ambient Global News Ribbon

- **Enterprise Data Source:** Server-side parsing of standard RSS Feeds (Reuters, AP, Bloomberg).
- **3-Column Grid Behavior:** Spans the three bottom columns (`transform-origin: bottom`). Headlines are cached locally into an array. If offline, the loop cycles cached headlines endlessly. Operates strictly with `translate3d` at an ultra-low cinematic crawl speed to prevent eye fatigue.

### 2. World Clock Matrix

- **Enterprise Data Source:** Internal system clock with localized UTC offsets via JavaScript.
- **3-Column Grid Behavior:** Supports up to 9 time zones perfectly distributed across the 3x3 grid. Renders in a pure, non-overlapping HH:MM format. Responsive scaling keeps typographic baselines intact across all devices.

### 3. Advanced Weather & Astronomical HUD

- **Enterprise Data Source:** Open-Meteo API (fetched defensively every 5 mins) and local JavaScript-based solar/lunar algorithms (sunrise, sunset, 8 moon phases).
  - **3-Column Grid Behavior:** Grid items dynamically shift visual state based on `.card-day` and `.card-night` CSS classes. In *Observatory Mode*, it highlights subtle star map circles and moon phases at the absolute visual center.
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## CATEGORY 5: GENERATIVE ART & AMBIENT TELEMETRY (Aesthetic Focal Points)

Focus: Pure visual luxury and mathematical, low-CPU motion for architectural walls.

### 1. Generative Horology Mode

- **Enterprise Data Source:** Internal system time mapped to a local mathematical physics engine (spring-damper models).
- **3-Column Grid Behavior:** Occupies the center column. Replaces digital numbers with concentric, ultra-thin rings (hours, minutes, seconds). Uses a `requestAnimationFrame` canvas loop with parametric equations  $(x(\theta)=(R+r \cdot \sin(\theta) \cdot \cos(\theta)))$  to create fluid, sweeping organic paths without mechanical ticking.

### 2. Neural Network Traffic Shimmer

- **Enterprise Data Source:** On-site Crestron PIR/Ultrasonic foot-traffic density sensors via BACnet, translated by the backend.
- **3-Column Grid Behavior:** Spans across columns. Maps real-time foot traffic volume into the base velocity and turbulence of a low-CPU vector particle field. The primary color temperature shifts based on the time of day, going from warm copper to deep charcoal.

### 3. Micro-Climate Particle Overlay

- **Enterprise Data Source:** WeatherFlow Tempest hardware (pyranometer, anemometer) over local UDP (port 50222) or Cloud REST API.
- **3-Column Grid Behavior:** Acts as a master visual modifier over the grid. Avoids CPU-heavy canvas weather storms. Uses pure GPU compositing: *Storm Dimming* uses infinite `ease-in-out` opacity pulsing; *Rain Reflection Shimmer* uses animated `linear-gradient` pseudo-elements over the grid containers; *Wind Vector* dynamically alters the rotation angle ( $\theta$ ) of the module's graphics.